

Very compact external cavity diffraction-coupled tapered laser diodes

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Received 12 October 2007; accepted 4 January 2008;
posted 11 January 2008 (Doc. ID 88566); published 15 February 2008

To maintain the same beam quality as that of a single emitter and to be close to diffraction limit, we have combined a phase corrected array, emitting at $\lambda = 975$ nm, coherently using the Talbot effect. First, to improve the beam quality of the array, a phase correcting system was added. The FWHM divergences of the array (which is approximately the same as that of the single emitter since the emitters within the array are not optically coupled to each other) were reduced from 34° to 0.17° in the fast axis and from 3.5° to 0.7° in the slow axis at 6 A. Then, to be close to the diffraction limit, we have combined this corrected array coherently using the Talbot effect. We have obtained a quasi-monolobe slow axis far field profile for the in-phase mode with a central peak divergence of only 0.27° at 1.5 A, 315 mW under cw operation, and of only 0.20° at 2.5 A, 787 mW under pulsed operation. To our knowledge, this is the first demonstration of coherent coupling of a corrected tapered laser diode array in an external Talbot cavity. © 2008 Optical Society of America

OCIS codes: 140.2010, 140.0140, 140.5960.

1. Introduction

Tapered lasers offer both high power, together with good beam quality. They contain a ridge waveguide acting as a modal filter and a tapered section of increasing width, providing high power. We have realized arrays of several emitters (N), which are not optically coupled to each other. Consequently, the beam quality parameter (M^2) of the array equals at least N times the single emitter one.

For improving the beam quality, an acylindrical microlens has been added for collimating the fast axis, and a phase plate makes it possible collimating the slow axis and correcting the residual aberrations

of the microlens and beam pointing error. With this correcting system, the FWHM divergences were reduced from 34° to 0.17° in the fast axis and from 3.5° to 0.7° in the slow axis at 6 A.

To maintain the same beam quality as that of a single emitter and to be close to the diffraction limit, we have investigated to combine this corrected array coherently using the Talbot effect. We have obtained a quasi-monolobe slow axis far field profile for the in-phase mode with a central peak divergence of only 0.18° at 900 mA, which leads to a quasi-circular beam. The peak divergence is 0.27° at 1.5 W, 315 mW under cw operation and only 0.17° at 2 A, 563 mW under pulsed operation. To our knowledge, this is the first demonstration of coherent coupling of a corrected tapered laser diode array in an external Talbot cavity.

2. Laser Structure

Our lasers are based on an Al-free active region, which is favorable for good reliability. The material structure was grown by metallorganic chemical vapor deposition on a GaAs substrate. The active layer consists of a strained GaInAs quantum well embedded in a large optical cavity (LOC). GaInAsP and GaAlAs alloys are used as optical confinement and cladding layers, respectively (Fig. 1). This structure exhibits low internal losses of 3 cm^{-1} , a very low transparency current density of 85 A/cm^2 , and an excellent internal quantum efficiency of 95%.

Based on these good results, fully index-guided (IG) tapered lasers emitting at $\lambda = 975\text{ nm}$ [1] have been realized using a ridge structure with a narrow taper angle of less than 1° and a total cavity length of 2.5 mm (Fig. 2). The devices have a narrow output width of a few tens of micrometers. The rear and the output facets are highly reflective (HR) and partially reflecting coated, and they are mounted episcide down with indium. This laser delivers a maximum power of 1 W cw, together with a good beam quality parameter $M^2 = 3.5$.

3. Tapered Laser Array with Phase Correction

For reaching higher power, we have realized an array of $N = 10$ IG emitting at $\lambda = 975\text{ nm}$ (Fig. 3). The spacing between the emitters is $d = 100\text{ }\mu\text{m}$.

This array has a threshold current of 945 mA , an external differential efficiency of 0.9 W/A , and a high wall plug efficiency of 48%. It delivers 3.6 W at 6 A 20°C cw. The bar was first collimated in the fast axis, using an acylindrical lens. A fused silica phase plate was used to correct the wavefront of the bar (Fig. 4). Wavefront reconstruction was used to calculate the optical surface of this phase plate. This approach provides an optimal collimation for each individual emitter [2]. With this correcting system, the FWHM divergences were reduced from 34° to 0.17° in the fast axis and from 3.5° to 0.7° in the slow axis at 6 A , and the individual beam quality parameter is of only 1.35 (Figs. 5 and 6).

4. Coherent Coupling in an External Talbot Cavity

In order to improve beam quality of diode arrays, several approaches have been investigated to combine them coherently such as evanescent coupling [3], intracavity spatial filtering [4], or a combining technique using a binary phase grating [5] and also the Talbot effect. For the Talbot effect, both monolithic [6] as well as external Talbot cavity [7] configurations

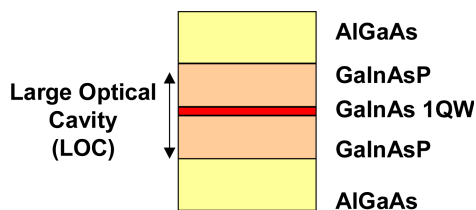


Fig. 1. (Color online) Al-free active region laser structure.

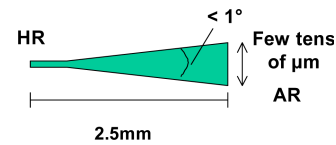


Fig. 2. (Color online) Geometry of the index-guided tapered laser.

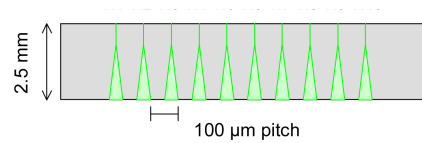


Fig. 3. (Color online) Geometry of the array of ten tapered lasers.

have been demonstrated. The Talbot effect refers to a diffraction phenomenon and consists of a reproduction of the field of an illuminated periodic object at certain distances away from the object plane. These distances are multiples of the Talbot distance $Z_T = 2d^2/\lambda$, where d is the spatial periodicity of the object and λ the wavelength. It was studied for many kinds of lasers such as CO_2 laser [8] or solid state lasers [9]. Particular interest was placed on semiconductor lasers because of their small size and high efficiency.

By using a feedback element, we can superimpose the image plane with the object one and induce coherent coupling between the lasers. In this work, the output mirror with a reflectivity of 50% is at a distance $Z_T/4 = 0.5\text{ cm}$ from the array. The distance after one round trip is $Z_T/2$. This Talbot setup normally selects the out-of-phase supermode, in which the phase of each emitter is opposite to that of its neighbor. When the mirror is tilted by an angle $\alpha = \lambda/2d$, the in-phase supermode is selected [10].

5. Experimental Results

A. Power Characteristics

We have measured the power characteristics for the array of 10IG1 in the external Talbot cavity for the in-phase and the out-of-phase modes. We have obtained approximately the same power characteristic for both of them.

In Talbot cavity, we have obtained a lower threshold current of 730 mA and a lower external efficiency of 0.39 W/A (Fig. 7) in comparison with the threshold current of the free running device of 945 mA and its



Fig. 4. (Color online) Corrected array of tapered laser diodes.

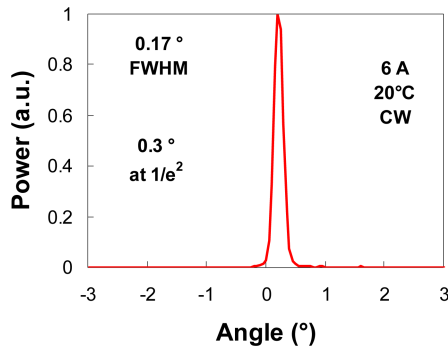


Fig. 5. (Color online) Vertical far field profile of the corrected array.

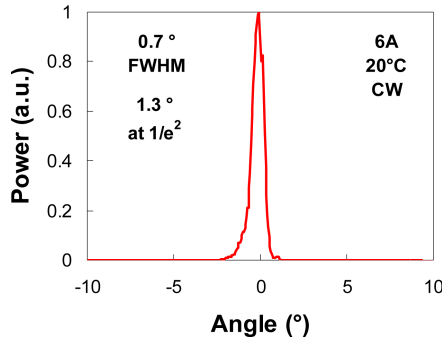


Fig. 6. (Color online) Horizontal far field profile of the corrected array.

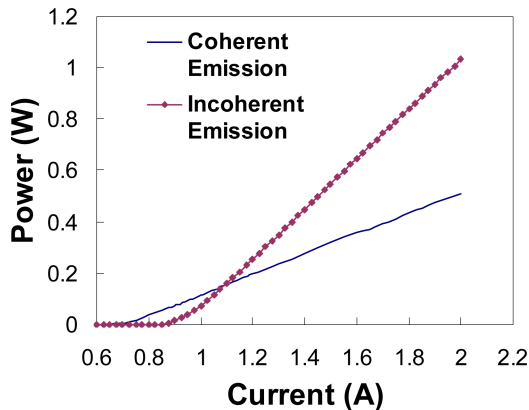


Fig. 7. (Color online) Power characteristics of the incoherent and coherent emissions under cw operation.

external efficiency of 0.9 W/A. This is due to the combined effect of the higher reflectivity of the output mirror (50%) in comparison with that of the free running devices (5%) and the additional losses of the Talbot cavity of 0.1 cm^{-1} .

B. Far Field Profile under cw Operation

First, the far field profile was recorded near threshold at 900 mA when the output mirror is not tilted (Fig. 8). It has a total aperture of 0.67° FWHM, which is as large as that of the free running device (0.7° FWHM), and is modulated by diffraction peaks.

The two highest peaks indicate the out-of-phase supermode. Their divergence equals only 0.19° , which is 3.4 times the diffraction limit of $\lambda/(Nd)$.

The output mirror was then tilted by $\alpha = \lambda/2d$ (Fig. 9). The highest peak is centered, which indicates the in-phase supermode. Its measured width is of 0.18° FWHM. With the FWHM divergence of 0.17° in the fast axis, we obtain a circular beam.

Figures 10 and 11 show the same measurement at higher drive current (1.5 A, 315 mW). We have noticed that the peak divergences become larger, with 0.27° for the in-phase mode and 0.32° for the out-of-phase mode, and that the intensity of the incoherent background is higher. This is the result of the competition between coherent modes created by the Talbot cavity and the incoherent modes emitted by the individual lasers. Indeed, the output facet of the array we use now is coated for maximum output power without Talbot cavity. This allows self-oscillation of

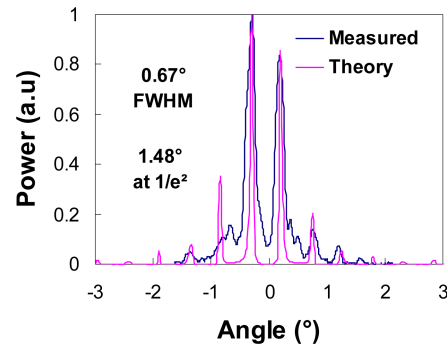


Fig. 8. (Color online) Far field at 900 mA (no tilt).

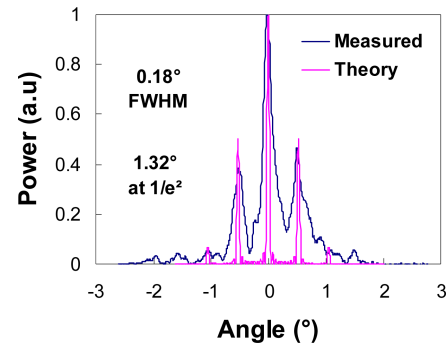


Fig. 9. (Color online) Far field at 900 mA ($\alpha = \lambda/2d$).

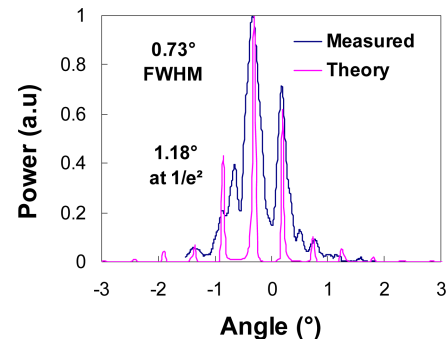


Fig. 10. (Color online) Far field at 1.5 A (no tilt).

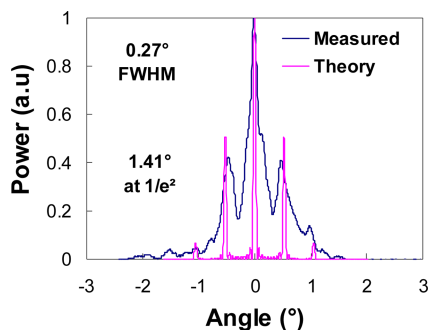


Fig. 11. (Color online) Far field at 1.5 A ($\alpha = \lambda/2d$).

the individual lasers and decreases the coupling efficiency between lasers for the coherent modes.

C. Far Field Profile under Pulsed Operation

Other measurements were carried out under pulsed operation (pulse width = 400 ns and repetition rate = 1 kHz), to dissociate two important phenomena in semiconductor lasers; the thermal effect and the spatial hole burning. Under pulsed operation, the thermal effect is negligible, and at the same current of 1.5 A, 355 mW, we have obtained narrower divergences of 0.15° FWHM (which is also the FWHM peak divergence) and 1.18° at $1/e^2$, in comparison with the continuous wave operation for the in-phase mode (Fig. 12). The coherence is maintained at a higher current of 2 A, 563 mW with a FWHM peak divergence of only 0.17° and 1.19° at $1/e^2$ (Fig. 13), but the incoherent background is still con-

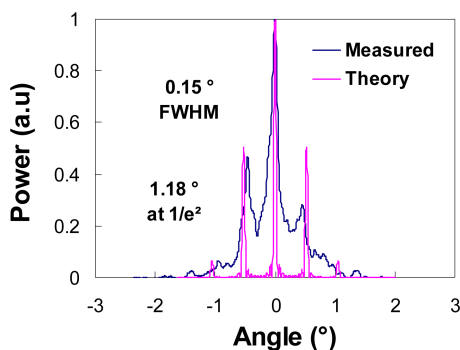


Fig. 12. (Color online) Far field at 1.5 A (pulsed operation, in-phase mode).

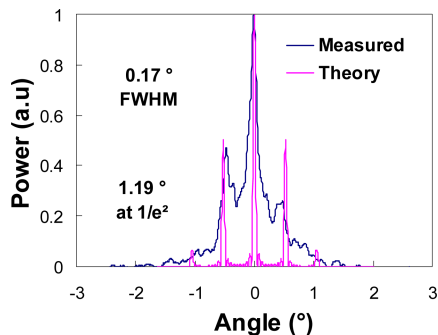


Fig. 13. (Color online) Far field at 2 A (pulsed operation, in-phase mode).

siderable at currents larger than 2.5 A, 787 mW with a FWHM peak divergence of 0.20° and a $1/e^2$ divergence of 1.34° (Figs. 14 and 15).

For the out-of-phase mode (Figs. 16–19) the peaks are narrower in comparison with the cw operation,

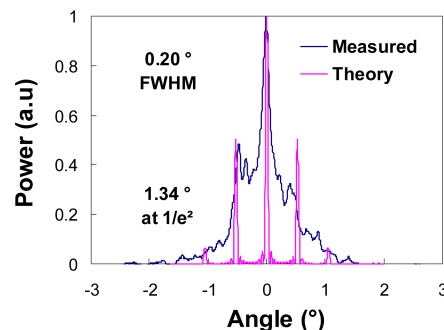


Fig. 14. (Color online) Far field at 2.5 A (pulsed operation, in-phase mode).

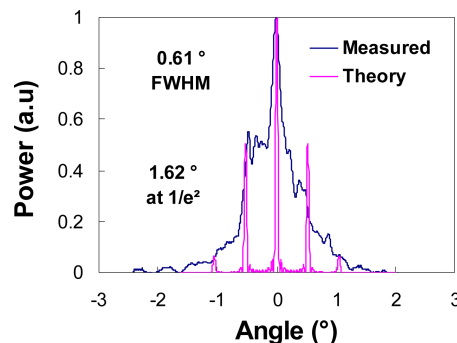


Fig. 15. (Color online) Far field at 3 A (pulsed operation, in-phase mode).

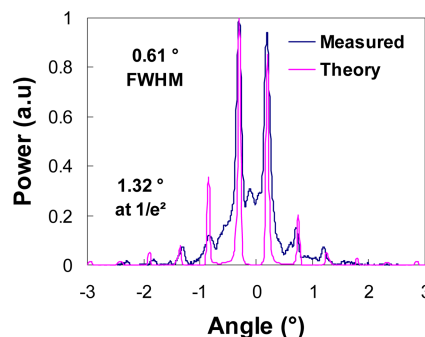


Fig. 16. (Color online) Far field at 1.5 A (pulsed operation, out-of-phase mode).

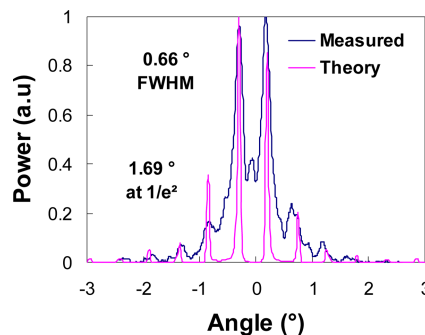


Fig. 17. (Color online) Far field at 2 A (pulsed operation, out-of-phase mode).

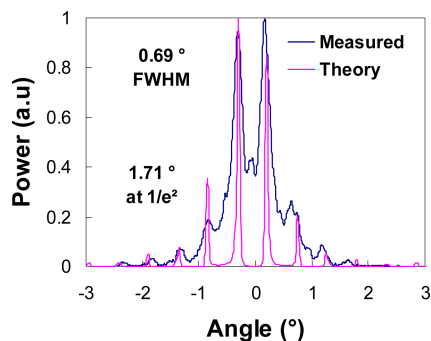


Fig. 18. (Color online) Far field at 2.5 A (pulsed operation, out-of-phase mode).

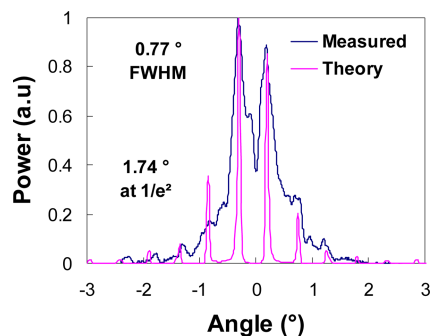


Fig. 19. (Color online) Far field at 3 A (pulsed operation, out-of-phase mode).

with a FWHM peak divergence of only 0.12° at 1.5 A, 0.18° at 2 A, and 0.20° at 2.5 A, but the incoherent background is considerable at currents larger than 2.5 A, 787 mW.

6. Conclusion

We have demonstrated for the first time the in-phase operation in an external Talbot cavity of a linear array of ten tapered laser diodes that has been corrected with both fast-axis collimation using an acylindrical microlens and slow-axis collimation using a laser written phase plate. The FWHM diver-

gences are 0.17° and 0.27° , respectively, in the fast and slow axis at 1.5 A, 315 mW under cw operation. We have obtained narrower divergences under pulsed operation with a central peak divergence of only 0.20° at 2.5 A, 787 mW.

The authors gratefully acknowledge the partial support of the European Commission through the IST project 035-266, BRIGHTER. We also thank Y. Robert, M. Carbonnelle, and C. Dernazerat for excellent technical assistance.

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